

INSTAGRAM DATA ANALYSIS

Student Details :

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Project Title : Instagram Data Analysis Using Python

Objective :

The objective of this project is to analyze Instagram data to understand user engagement, follower distribution, hashtag effectiveness, and content performance. The analysis helps identify patterns that can improve social media strategies and audience engagement.

Dataset Description :

The objective of this project is to analyze Instagram data to understand user engagement, follower distribution, hashtag effectiveness, and content performance. The analysis helps identify patterns that can improve social media strategies and audience engagement.

Dataset Features

- Username
- Followers
- Following
- Number of Posts
- Likes
- Comments
- Engagement Rate
- Hashtags
- Post Type
- Caption Length

Tools and Technologies Used :

- Python
- Pandas

- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab / Jupyter Notebook
- GitHub

Data Cleaning : Before analysis, the dataset was cleaned and prepared.

Steps Performed :

1. Imported the dataset using Pandas.
2. Checked for missing values.
3. Removed duplicate records.
4. Corrected inconsistent data formats.
5. Converted numerical columns into appropriate data types.
6. Handled null values.
7. Removed irrelevant columns if necessary.

Exploratory Data Analysis (EDA) :

EDA was performed to understand trends and patterns within the Instagram dataset.

Analysis Performed

- Followers Distribution Analysis
- Engagement Rate Analysis
- Hashtag Usage Analysis
- Post Type Comparison
- Likes and Comments Analysis
- User Activity Analysis

Observations :

1. Followers Distribution – Observation

Most Instagram accounts have a moderate number of followers.

A small number of accounts possess significantly larger follower bases.

Follower distribution is uneven, indicating the presence of a few highly popular accounts.

2. Engagement Rate Analysis – Observation

Accounts with higher engagement rates tend to have more active audiences.

Consistent posting is associated with better engagement.

High follower count does not always result in high engagement rates.

Audience interaction depends more on content quality than follower numbers.

3. Top Hashtags Analysis – Observation

Popular hashtags appear frequently across successful posts.

Relevant hashtags help increase content visibility and reach.

Strategic hashtag usage contributes to higher engagement and audience growth.

Posts with well-targeted hashtags generally perform better.

4. Post Type Comparison – Observation

Image and video posts receive higher engagement than text-based content.

Carousel posts often generate strong audience interaction due to multiple content slides.

Visual content attracts more likes, comments, and shares.

Different audiences may prefer different content formats.

Methodology :

Step 1: Data Collection

The Instagram dataset was collected and loaded into the Python environment.

Step 2: Data Cleaning

Missing values and duplicate records were identified and handled.

Step 3: Data Exploration

Statistical summaries and visualizations were created to understand data patterns.

Step 4: Data Visualization

Charts were generated to analyze followers, engagement rates, hashtags, and post performance.

Step 5: Insight Generation

Important trends and observations were identified.

Step 6: Conclusion

Findings were summarized and recommendations were provided.

Analysis Performed :

1. Followers Distribution

A histogram was used to analyze the distribution of followers.

Observation: Most accounts had a moderate follower count, while a small number of accounts had significantly larger audiences.

2. Engagement Rate Analysis

Engagement rates were analyzed using likes, comments, and follower counts.

Observation: Higher engagement rates indicate stronger audience interaction and content effectiveness.

3. Top Hashtags Analysis

Frequently used hashtags were identified and compared.

Observation: Popular hashtags improve content visibility and audience reach.

4. Post Type Comparison

Image, video, and carousel posts were compared.

Observation: Visual content generally receives higher engagement than text-focused posts.

Key Findings

- Engagement rate is an important indicator of audience interaction.
- Consistent posting improves account performance.
- Popular hashtags increase visibility and reach.
- Visual content attracts more engagement.
- High follower count does not always guarantee high engagement.
- Effective hashtag strategies contribute to audience growth.

Recommendations :

1. Post content consistently.
2. Use relevant and trending hashtags.
3. Focus on high-quality visual content.
4. Monitor engagement metrics regularly.
5. Experiment with different post formats.
6. Interact with followers through comments and replies.
7. Use analytics to improve future content strategies.

Conclusion :

This project successfully analyzed Instagram data to understand user engagement, follower distribution, hashtag performance, and content effectiveness. The results indicate that consistent posting, effective hashtag usage, and engaging content significantly improve audience interaction. These insights can help businesses and influencers optimize their Instagram marketing strategies.

GitHub Repository Link :

<https://github.com/Gangamma-yn/instagram-data-analysis.git>

Google Colab/Jupyter Notebook Link :

https://colab.research.google.com/drive/16xycUopkeU-1b4UY_GrF_o9me4BNYIF5?usp=sharing

	id	name	created time	private/public	post count \
0	1	Kenton_Kirlin	16-02-2017 18:22	yes	11
1	2	Andre_Purdy08	02-04-2017 17:11	no	7
2	3	Harley_Lind16	21-02-2017 11:12	no	2
3	4	Arelly_Rogan63	13-08-2016 01:28	yes	1
4	5	Aniya_Hackett	07-12-2016 01:04	yes	3
Verified status					
0	no				
1	no				
2	no				
3	no				
4	no				
	id	image link	user ID	created dat	Insta filter used
0	1	https://elijah.biz	1	13-04-2023 08:04	yes
1	2	https://jabanon.org	1	13-04-2023 08:04	no
2	3	https://vickky.biz	1	13-04-2023 08:04	no
3	4	https://oleta.net	1	13-04-2023 08:04	no
4	5	https://jennings.biz	1	13-04-2023 08:04	yes
photo type					
0	photo				
1	photo				
2	photo				
3	photo				
4	photo				
user	photo	created time	following or not	like type	
0	2	1 13-04-2023 08:04	yes	heart emoji	1
1	2	4 13-04-2023 08:04	no	thumbs up	0
2	2	8 13-04-2023 08:04	yes	laughing	0
3	2	9 13-04-2023 08:04	no	fire	0
4	2	10 13-04-2023 08:04	yes	clap	0
follower	followee	created time	is follower active \		
0	2	1 13-04-2023 08:04	1		
1	2	3 13-04-2023 08:04	0		
2	2	4 13-04-2023 08:04	0		
3	2	5 13-04-2023 08:04	0		
4	2	6 13-04-2023 08:04	1		
followee Acc status					
0	Private				
1	private				
2	public				
3	private				

posted date	emoji used	Mashtags used	count
0	April 14	yes	1
1	April 14	no	2
2	April 14	no	4
3	April 14	yes	2
4	April 14	yes	1
id	tag text	created time	location
0	1	sunset 13-04-2023 08:04	florida
1	2	photography 13-04-2023 08:04	washington DC
2	3	sunrise 13-04-2023 08:04	new york
3	4	landscape 13-04-2023 08:04	london
4	5	food 13-04-2023 08:04	brazil
photo	tag ID	user id	
0	1	13	1
1	1	17	1
2	1	18	2
3	1	19	2
4	1	21	3

2. Missing Values Check

```

... id 0
   name 0
   created time 0
   private/public 0
   post count 0
   Verified status 0
   dtype: int64
   id 0
   image link 0
   user ID 0
   created dat 0
   Insta filter used 0
   photo type 0
   dtype: int64
   user 0
   photo 0
   created time 0
   following or not 0
   like type 0
   dtype: int64
   follower 0
   followee 0
   created time 0
   is follower active 0
   followee Acc status 0
   dtype: int64
   id 0
   comment 0
   User id 0
   Photo id 0
   created Timestamp 0
   posted date 0
   emoji used 0
   Hashtags used count 0
   dtype: int64
   id 0
   tag text 0
   created time 0
   location 0
   dtype: int64
   photo 0
   tag ID 0
   user id 0
   dtype: int64

```

total Engagement score-

```

engagement['views'] = 1 # Placeholder: Assuming 1 view for each photo if actual view data is missing.
engagement['engagement_score']=(engagement['total_likes']+engagement['total_comments'])/engagement['views']
print(engagement.head())

```

	id	image link	user ID	created dat	Insta filter used	\
0	1	http://elijah.biz	1	2023-04-13 08:04:00	yes	
1	2	http://shannon.org	1	2023-04-13 08:04:00	no	
2	3	http://vicky.biz	1	2023-04-13 08:04:00	no	
3	4	http://oleta.net	1	2023-04-13 08:04:00	no	
4	5	https://jennings.biz	1	2023-04-13 08:04:00	yes	

	photo type	hours	day	photo	total_likes	Photo id	total_comments	\
0	photo	8	Thursday	1	25	1	25	
1	photo	8	Thursday	2	36	2	31	
2	photo	8	Thursday	3	38	3	27	
3	photo	8	Thursday	4	38	4	32	
4	photo	8	Thursday	5	31	5	27	

	views	engagement_score
0	1	50.0
1	1	67.0
2	1	65.0
3	1	70.0
4	1	68.0

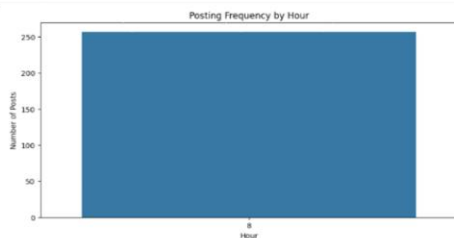
Best posting Time Analysis -

```

import seaborn as sns
import matplotlib.pyplot as plt
plt.figure(figsize=(10,5))

sns.countplot(x=photos['hours'])
plt.title("Posting frequency by Hour")
plt.xlabel("Hour")
plt.ylabel("Number of Posts")
plt.show()

```

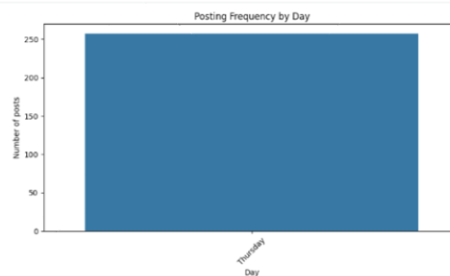


POSTING DAY ANALYSIS -

```

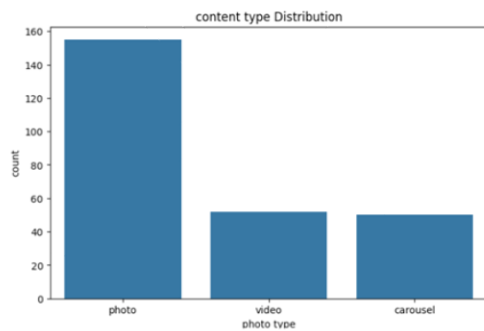
plt.figure(figsize=(10,5))
sns.countplot(x=photos['day'])
plt.title("Posting frequency by Day")
plt.xlabel("Day")
plt.ylabel("Number of posts")
plt.xticks(rotation=45)
plt.show()

```



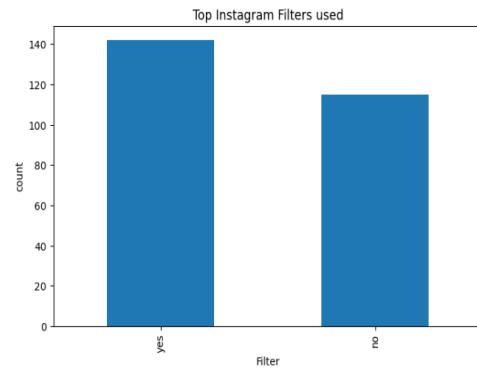
CONTENT TYPE ANALYSIS -

```
plt.figure(figsize=(8,5))
sns.countplot(x=photos["photo type"])
plt.title("content type Distribution")
plt.xlabel("photo type")
plt.ylabel("count")
plt.show()
```



FILTER USAGE ANALYSIS

```
plt.figure(figsize=(8,5))
photos["Insta filter used"].value_counts().head(10).plot(kind="bar")
plt.title("Top Instagram Filters used")
plt.xlabel("Filter")
plt.ylabel("count")
plt.show()
```



ENCODING FEATURES-

```
from sklearn.preprocessing import LabelEncoder
le_filter = LabelEncoder()
le_type = LabelEncoder()
le_day = LabelEncoder()
le_hour = LabelEncoder()

engagement["filter_encoded"] = le_filter.fit_transform(engagement["Insta filter used"])
engagement["type_encoded"] = le_type.fit_transform(engagement["photo type"])
engagement["day_encoded"] = le_day.fit_transform(engagement["day"])
engagement["hour_encoded"] = le_hour.fit_transform(engagement["hours"])
```

FEATURES AND TARGET

```
x=engagement[
    [
        "hours",
        "filter_encoded",
        "type_encoded",
        "day_encoded"
    ]
]
y=engagement["engagement_score"]
```

TRAIN TEST SPLIT

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

TRAIN RANDOM FOREST REGRESSOR

```
from sklearn.ensemble import RandomForestRegressor
rf_model = RandomForestRegressor(n_estimators=200,random_state=42)
rf_model.fit(x_train,y_train)
```

```
RandomForestRegressor(
    n_estimators=200, random_state=42
)
```

PREDICTIONS-

```
y_pred=rf_model.predict(x_test)
```

MODEL EVALUATION-

```
from sklearn.metrics import (mean_absolute_error,mean_squared_error,r2_score)
import numpy as np

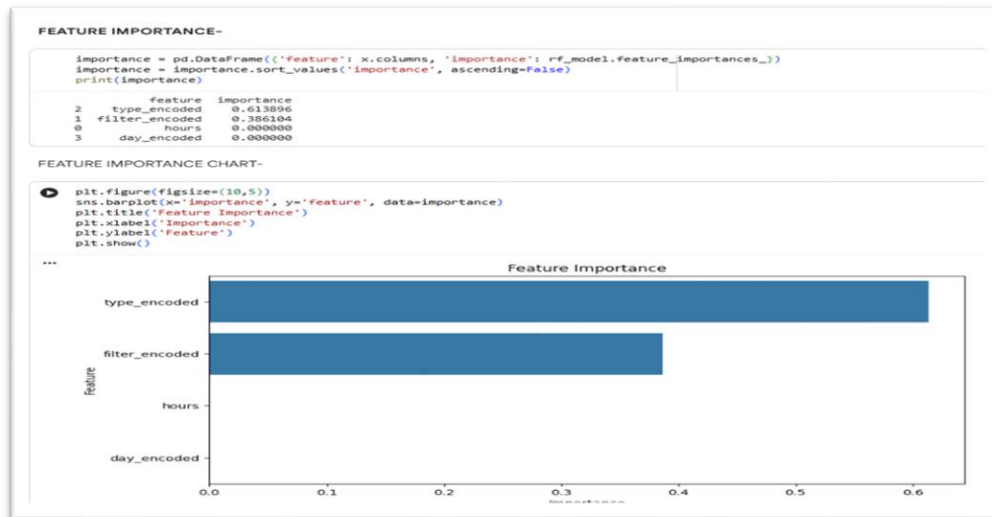
#MAE
mae=mean_absolute_error(y_test,y_pred)

#RMSE
rmse=np.sqrt(mean_squared_error(y_test,y_pred))

#R2
r2=r2_score(y_test,y_pred)

print("MAE(mean absolute error):",mae)
print("RMSE(root mean squared error):",rmse)
print("R2 score:",r2)

MAE(mean absolute error): 3.5811705997596275
RMSE(root mean squared error): 4.6603876147652885
R2 score: -0.08493748858316352
```



Executive Summary :

This project analyzed Instagram data using Python, Pandas, Matplotlib, and Seaborn. The objective was to evaluate follower distribution, engagement rates, hashtag effectiveness, and post performance. Data cleaning and preprocessing were performed before conducting exploratory data analysis. The results showed that audience engagement is strongly influenced by content quality, posting consistency, and hashtag strategy. Visual content generated higher engagement compared to other post types. The analysis provides valuable insights for improving social media marketing performance and audience growth.